Ethan Stark

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EDUCATION

Honours Bachelor of Science, University of Toronto **CGPA: 3.89/4.0**
Specialist in Computer Science, Minor in Mathematics **Expected Dec. 2026**

- *Achievements*: 2023 — 2025 Dean's List Scholar
- *Extracurriculars*: Math outreach volunteer at The Fields Institute
  - Designed and taught enrichment lessons for groups of **20–30 high school students**, translating unfamiliar mathematical concepts into accessible activities and facilitating collaborative problem-solving.
  - Supported Canadian Open Math Challenge grading for **6,000+ submissions** and invigilated the Kangaroo mathematics contest.
- *Relevant Coursework*: Machine Learning, Artificial Intelligence, Software Design, Databases, Web Programming, Systems Programming, Operating Systems, Algorithms & Data Structures, Probability & Statistics, Linear Algebra.

EXPERIENCE

Software Engineer Intern – AI Systems May 2025 — Aug. 2026
Huawei Canada Research Institute Markham, ON

- Engineered a modular, **low-latency Python** audio framework using concurrent threads and queues, real-time capture, playback, and routing for voice-assistant fine-tuning pipeline; validated reliability with automated tests.
- Built a synthetic audio generation pipeline using impulse-response convolution, producing **100+ hours of ML training data in under 5 minutes** and replacing hundreds of hours of manual collection.
- Developed a Kotlin Android application which integrates ONNX model inference, multichannel audio streaming, and asynch processing; helped take a voice-assistant prototype through **deployment on production devices**.
- Collaborated with engineers and researchers through code reviews, documentation, technical presentations, and investigation of complex model, software, and infrastructure issues.

PROJECTS

Scriptorium – Online Code Execution Platform *TypeScript, Node.js, Prisma, SQLite, Docker*
Project Link: `https://github.com/ethanrstark/scriptorium`

- Built a full-stack code execution and content-sharing application in a **three-person team**; designed a **relational database schema** for users, posts, comments, and execution records.
- Developed and tested **REST APIs** for CRUD operations, search and pagination, and database queries using Prisma and SQLite; integrated backend services with the React frontend and **Docker-based** program execution.

User-Level Threading Library *C, Linux, GDB*

- Implemented a **user-level threading library in C on Linux**, including thread creation, lifecycle management, context switching, and cooperative and preemptive scheduling. Debugged concurrency and memory issues and validated scheduler using targeted tests and **GDB** debugging.
- Developed **preemptive scheduling using timer signals**, disabling signals during critical thread-management operations to prevent race conditions. Built mutex locks for safe concurrent execution.

Marketplace App – Java/Spring Boot *Java, Spring Boot, REST APIs*
Project Link: `https://github.com/ethanrstark/scamazon`

- Collaborated in a **seven-person team** to develop an online marketplace using **object-oriented design**, SOLID principles, design patterns, and Clean Architecture; implemented user profiles and reviews.
- Separated business logic from UI and infrastructure to improve testability and maintainability; currently modernizing the application toward a **Spring Boot backend with REST APIs**.

SKILLS

- **Languages:** Python, Java, TypeScript, JavaScript, SQL, C++, C, Kotlin
- **Backend:** Node.js, REST APIs, Prisma, PostgreSQL, React
- **AI/ML:** PyTorch, NumPy, Pandas, Hugging Face, vLLM, ONNX, OpenAI API
- **Systems:** Linux, multithreading, concurrency, processes, networking, TCP/IP, filesystems, benchmarking
- **Tools:** Git/GitHub, Linux, Docker, Postman, AI-assisted development tools